

# Hyeonseop Yoon

Applied NLP / AI Researcher · Grounded Retrieval, Agent Systems, Evaluation

Seoul, Korea · xianxie31@korea.ac.kr · pfsv.github.io · GitHub · ORCID

## PROFILE

AI research engineer building grounded, evidence-driven language systems under production constraints. Experience spans retrieval and reranking, query expansion, RAG and agent systems, LLM adaptation, evaluation, and on-premise serving. Earlier research connected NLP representations with human cognition and neural signals.

## PROFESSIONAL EXPERIENCE

MAUM.AI

Mar 2024 — Present

AI Research Engineer · AICC R&D;

- Lead research and engineering across retrieval, RAG, agent orchestration, model adaptation, evaluation, and serving for automotive, insurance, finance, and public-sector systems.
- Developed a cascaded reranking and domain-adaptation pipeline that improved documented FAQ accuracy from 69.4% to 96.3% on a client evaluation set.
- Built verified-unit QA and staged linguistic seeding for two contact-center domains; the resulting application study was accepted to GroundLM at EMNLP 2026.
- Built structure-aware retrieval for constrained policy and technical documents, improving an internal exact-match evaluation from 4/59 to 47/59.
- Delivered on-premise LLM and RAG systems, including model fine-tuning, quantization, synthetic-data generation, offline deployment, distributed serving, and observability.

Seoul National University

Dec 2023 — Feb 2024

Researcher · Cognitive & Systems Neuroscience Laboratory

- Reviewed and analyzed RNN models of visual working memory, sensory encoding, and decision bias; inspected training dynamics and visualized unit-level information flow.

Korea University

Aug 2022 — Sep 2023

Researcher · Brain Signal Processing Laboratory

- Fine-tuned transformer and recurrent language models for reasoning tasks and extracted hidden representations and attention patterns.
- Compared model representations with voxel-level fMRI signals using representational similarity analysis and whole-brain mapping.

## SELECTED PROJECTS

Production Grounded AI Systems

2024 — 2026

MAUM.AI · Automotive, insurance, rail, and public-sector deployments

- K-company automotive AICC: domain-specific FAQ retrieval, slot-filling agent, long-term session memory, model compression, and low-latency inference across multiple service locations.
- K-company rail service: on-premise voice chatbot and offline regulation-search agents using synthetic corpus generation and grounded retrieval.
- K-company insurance systems: document-layout parsing, policy-code mapping agents, knowledge distillation, retrieval over internal regulations, and distributed model serving.

pycag

2026

Open-source Cache-Augmented Generation toolkit

PyPI package and public source: [github.com/PFSV/cag](https://github.com/PFSV/cag)

Korean Retrieval Embeddings

2025 — 2026

BGE-M3 and Qwen3-based public encoders

Model cards document training scope, AutoRAG evaluation, and limitations.

VisionCardio

2025

On-device rPPG research prototype

PyTorch training, Core ML export, SwiftUI integration, and deterministic wellness coaching policy.

## PUBLICATIONS

[P4] Staged Linguistic Seeding: Grounded Query Expansion for Verified-Unit QA in AI Contact Centers. Hyeonseop Yoon, Jeong-Eun Park. GroundLM Workshop at EMNLP 2026. arXiv:2609.00844

[P3] Metaphor in Mind and Machine. Hyeonseop Yoon et al. OHBM 2024 / Aperture Neuro. DOI

[P2] Comparative Analysis of Brain and NLP Models for Reasoning Tasks. Hyeonseop Yoon et al. Brain Engineering Society of Korea, 2023.

[P1] Korean Twitter Bot Detection based on Deep Learning. Hyeonseop Yoon et al. Korea Software Congress, KIISE, 2022.

ACADEMIC SERVICE

|                                                                                |      |
|--------------------------------------------------------------------------------|------|
| Reviewer                                                                       | 2026 |
| Grounding Language Models (GroundLM) Workshop at EMNLP                         |      |
| Reviewer                                                                       | 2026 |
| Vision-Language Models for Real-World Deployment (VLM4RWD) Workshop at NeurIPS |      |

SELECTED HONORS

|                                                                  |      |
|------------------------------------------------------------------|------|
| Outstanding Poster Award                                         | 2023 |
| Korean Society for Human Brain Mapping                           |      |
| Metaphor in Mind and Machine                                     |      |
| Outstanding Poster Award                                         | 2023 |
| Brain Engineering Society of Korea                               |      |
| Comparative Analysis of Brain and NLP Models for Reasoning Tasks |      |
| Outstanding Research Award                                       | 2022 |
| Korea Software Congress, KIISE                                   |      |
| Korean Twitter Bot Detection based on Deep Learning              |      |

ADDITIONAL EXPERIENCE & SKILLS

Open Source Contribution Academy Masters (Aug-Dec 2022) — Developed and maintained VisualPython, a Python-based GUI framework for statistics, machine learning, and deep learning workflows.

Data Youth Campus (Jul-Sep 2022) — Completed an NLP and deep-learning convergence program; built a Korean spam-detection system.

Technical: Python, PyTorch, Hugging Face, vLLM, RAG, BM25/SPLADE/dense retrieval, LoRA/ORPO, Docker, PostgreSQL/pgvector, Ray, SLURM/H100, Core ML, SQL, NumPy, pandas, scikit-learn

Language: Korean (native), English (TEPS 394, Feb 2024; lived in the United Kingdom, 2011–2013)

Military status: Not completed